Lehrstuhl für Data Science

Zero-Shot Narrative Classification via Agentic and Actor-Critic LLM Pipelines

Masterarbeit von

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Abstract

The task of assigning multiple, hierarchically-structured labels to text documents, known as Hierarchical Multi-Label Classification (HMLC), is critical in domains from scientific archiving to legal analysis. This review traces the methodological evolution of HMLC, beginning with foundational problem transformation methods like Binary Relevance and Classifier Chains, which primarily address the challenge of label correlation. We then examine the paradigm shift introduced by pre-trained Transformers, dissecting the dichotomy between local, top-down approaches prone to error propagation and global, hierarchy-aware models that integrate structural constraints via specialized loss functions or Graph Neural Networks. Finally, we explore the current frontier, where Large Language Models (LLMs) are reframing the task through generative paradigms, enabled by techniques such as LLM-powered data augmentation, instruction fine-tuning, and Parameter-Efficient Fine-Tuning (PEFT). This narrative highlights a progression towards increasingly sophisticated methods for embedding hierarchical prior knowledge into statistical models.

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1 Introduction

1.1 Motivation

Text Classification (TC) is a foundational task in Natural Language Processing (NLP) [Zan+24]. While early work often framed TC as a multiclass problem, modern texts regularly contain multiple overlapping themes [Hu+25; TS18]. This motivates Multi-Label Classification (MLC), where instances can be assigned multiple labels simultaneously [TS18]. Beyond mere multiplicity, labels are frequently correlated [TS18; Hua+24], so modeling inter-label dependencies is crucial.

A particularly compelling setting is hierarchical multi-label classification (HMLC), where labels are arranged in a predefined hierarchy (e.g., tree or DAG) from coarse to fine categories [Liu+23]. This structure matches how humans reason about complex discourse. In the high-stakes context of online news and crisis communication, identifying nested narratives is both socially relevant and technically challenging. As a further motivation, this work aims to rigorously evaluate the classification capabilities of large language models (LLMs) on a difficult, nuanced HMLC task—one that requires fine-grained distinctions, hierarchical consistency, and evidence-grounded reasoning rather than shallow keyword matching.

1.2 Problem Definition

We ground our study in SemEval-2025 Task 10, Subtask 2 [Pis+25], a hierarchical narrative detection task for news. The objective is to assign each document multiple labels from a two-level taxonomy of narratives and subnarratives, i.e., mapping a text to a set of (narrative, subnarrative) pairs. The task features: (i) extreme class imbalance; (ii) hierarchical consistency constraints; and (iii) multilingual complexities. Formally, the MLC problem seeks to construct a function $f : X \to 2^L$ that maps an instance $x \in X$ to a subset $Y \subseteq L$, where L is the finite label set [TS18]. This function may be learned through supervised training or constructed via other means (e.g., zero-shot prompting). HMLC adds the requirement that predicted labels respect the hierarchical structure [Liu+23].

1.3 Related Work

Classical MLC methods. Foundational problem-transformation techniques include Binary Relevance (BR) [Zha+18b], Classifier Chains (CC) [LDH24; Wen+20], and Label Powerset (LP) [Sha+18]. While simple and effective in some settings, these approaches can struggle to exploit complex dependencies and often require substantial labeled data.

Deep learning and Transformers. The advent of Transformer-based models, such as BERT [Dev+19] and XLM-RoBERTa [Con+20], established strong baselines for MLC by learning contextual representations that capture label correlations implicitly. However, in data-scarce and long-tailed regimes, fine-tuning can overfit to head classes and underperform on rare labels.

Hierarchy-aware architectures. Specialized HMLC methods incorporate the label hierarchy via hierarchical classifiers, structured losses, or graph-based modeling [SC22; Pen+21; Gon+20]. These approaches improve consistency across levels but typically remain supervised and sensitive to imbalance.

Zero-shot and LLM-based approaches. Recent practice leverages LLMs for zero-shot classification via prompt engineering, exploiting world knowledge and reasoning without task-specific fine-tuning [SSB25b; SSB25a]. Agentic decompositions and pipeline designs decompose complex decisions into simpler steps with validation. While promising for nuanced tasks, they can be sensitive to prompt phrasing, costly at inference, and require careful design to ensure traceability and reduce hallucination.

In summary, existing supervised baselines excel with sufficient balanced data but struggle in the long-tail, multilingual, and hierarchy-constrained settings of narrative detection. Zero-shot LLM pipelines offer an alternative by trading training for structured prompting and validation.

1.4 Our Contributions

We make the following contributions in the context of HMLC for narrative detection:

- We introduce a zero-shot agentic framework that decomposes narrative detection into specialized binary decisions handled by LLM agents, achieving competitive performance without task-specific fine-tuning.
- We propose a configurable, traceable actor - critic pipeline that augments the agentic approach with explicit evidence extraction and validation, improving robustness and interpretability.
- We provide a comparative analysis of three paradigms: a fine-tuned Transformer baseline, the agentic framework, and the actor - critic pipeline, highlighting trade-offs in accuracy, traceability, and cost.
- We conduct an in-depth error analysis on SemEval-2025 data, elucidating the roles of class imbalance, semantic ambiguity, and hierarchical constraints in model failures.

Paper outline. The remainder of this paper presents task and data analysis, details the system architectures, reports experiments and ablations, and concludes with discussion and future directions.

2 Related Work

2.1 Foundational paradigms of MLC

The main challenge in multi-label classification (MLC) was how to adapt algorithms designed for single-label (binary or multiclass) problems to handle multiple labels per instance. The core issue these methods faced was the presence of label correlations: in real-world data, labels are often not independent. For example, a news article tagged "Politics" is much more likely to also be tagged "Elections" than "Sports." Treating each label as a separate binary problem ignores these dependencies, potentially leading to suboptimal predictions.

Classical MLC methods can be seen as different strategies for balancing computational simplicity with the need to model label correlations. Some methods, like Binary Relevance, treat each label independently for simplicity, but this can miss important relationships between labels. Others, such as Classifier Chains or Label Powerset, explicitly model these correlations, but at the cost of increased computational complexity. The choice of method reflects a trade-off between efficiency and the ability to capture the true structure of the data.

2.1.1 Binary Relevance

Binary Relevance (BR) is the most intuitive approach. It decomposes the MLC problem with a label set of size $|\mathcal{L}|$ into independent binary classification problems. For each label, a separate classifier is trained to predict its presence or absence, effectively ignoring all other labels. [Zha+18b] The primary advantage of BR is its simplicity and efficiency, as the classifiers can be trained in parallel. Its main drawback, however, is the foundational label independence assumption, which completely disregards label correlations and can lead to lower predictive accuracy and logically incoherent label set predictions. [Suc+14]

2.1.2 Label Powerset

In contrast to decompositional methods, the Label Powerset (LP) method reframes the entire problem at once. It converts the multi-label task into a standard multi-class problem by mapping each distinct set of co-occurring labels to a single, unique class. For instance, the label sets 'Politics, Elections' and 'Sports, Weather' would become two separate classes for a multi-class classifier to learn. [Rea+11]

The main advantage of this approach is its ability to perfectly model the dependencies between labels for all combinations it has seen, as these correlations are baked into the class definitions [Suc+14]. However, this strategy is often impractical. The number of potential classes can become unmanageably large as the label set grows, a problem known as combinatorial explosion. This leads to a highly sparse class distribution where many label sets appear only a few times, making it difficult to train a robust model [CMM11]. Critically, the LP method cannot generalize to predict any combination of labels that did not appear in the training data.

2.1.3 Classifier Chains

The Classifier Chains (CC) method was proposed as a novel approach to overcome the stark trade-off between the label-independent BR and the computationally explosive LP. It seeks to model label dependencies while maintaining the efficiency of a binary relevance framework [Rea+11].

Like BR, CC trains $|\mathcal{L}|$ binary classifiers. However, instead of being independent, these classifiers are linked in a chain. The first classifier in the chain, $\mathcal{C}_1$, predicts the presence or absence of the first label. The predictions of this classifier are then used as additional features for the second classifier, $\mathcal{C}_2$, which predicts the second label. This process continues down the chain, with each classifier potentially benefiting from the predictions of all previous classifiers. This allows CC to capture label dependencies while still being relatively efficient to train. [Rea+11] The order of the chain can significantly impact performance, as earlier classifiers influence later ones [Rea+21]. To mitigate this, ensemble methods that average predictions over multiple random chain orders are often employed [Suc+14; Zha+18a].

2.2 Hierarchy aware architectures

The limitations inherent in classical problem transformation methods, particularly their struggles with large label spaces and their inability to deeply integrate structural information, precipitated a paradigm shift toward deep learning. We will focus on how neural architectures evolved from treating the label hierarchy as a post-hoc constraint to using it as a central component of the learning process. This evolution is characterized by a fundamental architectural divergence between local and global approaches, a conflict that was ultimately resolved through the powerful synthesis of Transformer-based text encoders and Graph Neural Networks (GNNs) for structure encoding.

2.2.1 From flat to hierarchical models

Early deep-learning methods for hierarchical multi-label classification often treated the task as a standard (flat) multi-label problem and ignored the label taxonomy. While these models learned strong text representations, they did not use the hierarchy's relationships. That matters because mistakes at higher levels of the taxonomy are semantically worse than small, nearby errors: for example, assigning a paper on "Quantum Mechanics" to "Arts" is much more serious than confusing "Physics" with "Chemistry." Flat models cannot distinguish these degrees of error. [Xu+21]

The paradigm shift occurred with the recognition that the hierarchy could be treated as a feature to guide learning, rather than just an output format. By making models "hierarchy-aware," it becomes possible to share statistical strength between parent and child nodes. For example, the few training instances available for a rare, specific label like "Superstring Theory" can be supplemented by the more abundant data from its parent labels, "String Theory" and "Theoretical Physics." This is especially crucial for improving performance on the long tail of infrequent labels that characterizes most real-world HMLC datasets. [Zan+24]

2.2.2 Local vs global approaches

The first generation of truly hierarchical models diverged into two main architectural philosophies: local and global. This division reflects a fundamental trade-off between capturing fine-grained, localized class relationships and maintaining a holistic, computationally tractable view of the entire label space.

2.2.3 Local approaches (Top down)

The local approach decomposes the hierarchical classification problem into a set of smaller, more manageable classification tasks distributed across the taxonomy. This is typically implemented as a top-down or "divide and conquer" strategy.

During inference, an instance is typically classified in a top-down manner. It is first evaluated by the classifier at the root; if a positive prediction is made for a node, the instance is then passed down to the classifiers of its children, and this process continues until a leaf node is reached or no further positive predictions are made. [RFR22]. While this approach excels at capturing the specific features that distinguish between closely related sibling classes, it suffers from a critical problem: **error propagation.** A single misclassification at a higher level of the hierarchy can irreversibly steer the prediction down an incorrect path, making it impossible to classify the instance into its correct, more specific sub-categories. [WCB18]

2.2.4 Global approaches (single classifier)

In contrast, the global approach uses a single, unified model to predict all labels in the hierarchy simultaneously. This is typically framed as a large multi-label classification problem where the output layer corresponds to the entire set of labels in the taxonomy. [WCB18]

The primary advantage of the global approach is that it inherently avoids the error propagation problem of local models, as all decisions are made in parallel by a single classifier. This makes the model more robust to errors at higher levels. Furthermore, global models are often more computationally efficient, as they require training only one model instead of a potentially large cascade of local classifiers. However, early global models faced a significant challenge: they struggled to effectively incorporate the complex structural information of the entire hierarchy into a single model. By treating the problem as a flat multi-label task, they often failed to capture the nuanced, local distinctions between sibling classes and could underfit the hierarchical relationships, thereby losing the very information that hierarchical classification aims to exploit. [WCB18; Zho+20]

2.2.5 Encoding the Hierarchy with Transformers and Graph Neural Networks

The solution to the local-versus-global problem came from combining two methods: Transformer models to understand text, and Graph Neural Networks (GNNs) to understand structure. This mix made it possible to build global models that are both efficient and aware of hierarchies. [WDH24; Zho+20]

GNNs are especially useful for working with label hierarchies. [Li+21] We can represent the taxonomy as a graph, where labels are nodes and parent-child links are edges. A GNN learns label representations by passing messages between connected nodes. Each node gathers information from its parents, children, and siblings. This way, the final label embeddings capture not just their own meaning, but also their place and relationships within the whole taxonomy.

A seminal work in this domain is the Hierarchy-Aware Global Model (HiAGM) by Zhou et al. (2020). HiAGM provides a blueprint for this new class of models. Its architecture consists of two main components: [Zho+20]

- A Text Encoder (e.g., BERT, RoBERTa) that generates a powerful contextualized representation of the input document.
- A Hierarchy-Aware Structure Encoder (e.g., a Bidirectional Tree-LSTM or a specialized Hierarchy-GCN) that operates on the label graph to produce hierarchy-aware label embeddings. This encoder models dependencies in both a top-down and bottom-up fashion, allowing information to flow in both directions across the hierarchy.

HiAGM further proposes two distinct fusion variants to combine text and structure features depending on the inference regime and efficiency/inductivity trade-offs.

HiAGM-LA (Multi-Label Attention) An inductive approach that uses a multi-label attention mechanism to compute document-specific label representations by attending from the text encoder outputs to the hierarchy-aware label embeddings. Because the attention weights are computed per document, the model can generalize to unseen instances without explicitly re-running a graph propagation step for each input, label embeddings can be pre-computed and stored, and the attention operation is applied at inference time.

HiAGM-TP (Text Feature Propagation) A deductive approach that propagates text features across the label hierarchy via graph propagation: document features are attached to label nodes and a GNN is run to diffuse these features through the taxonomy. This requires executing the GNN at inference time for each instance (or batch), which can be more computationally expensive but allows richer instance-specific propagation of evidence through the hierarchy.

The principles demonstrated by HiAGM have been extended and refined in subsequent work. For instance, Xu et al. (2021) proposed a framework using a loosely coupled GCN to explicitly model not only the vertical correlations (parent-child dependencies) but also the horizontal correlations (relationships between sibling nodes at the same level). This allows the model to capture, for example, that "Computer Science" and "Electrical Engineering" are more closely related to each other than to "History," even if they share the same parent 'Science & Technology'.

2.3 The rise of Large Language Models

The advent of Large Language Models (LLMs) has initiated another profound shift in the NLP landscape. Their advanced reasoning and generation capabilities are redefining their role in the HMLC pipeline, moving them beyond being simple end-point classifiers to becoming integral co-pilots in various stages of the workflow.

2.3.1 Zero shot and few-shot classification

The most immediate impact of LLMs is their remarkable ability to perform classification tasks with little to no task-specific training data. Through in-context learning, an LLM can be prompted with a task description and a few examples (few-shot) or even no examples (zero-shot) and perform hierarchical multi-label classification. This capability represents a paradigm shift for low-resource scenarios, potentially obviating the need for extensive data collection and annotation efforts; for narrative detection, one can provide definitions of the target narratives and ask the LLM to classify a new document accordingly. [Wan+23]

For instance, Eljadiri and Nurbakova (2025) demonstrate a high-performing zero-shot agentic approach to narrative classification. Their system instantiates multiple specialized LLM agents, each responsible for a binary decision on a single narrative or subnarrative label, while a meta-agent aggregates the binary outputs into final multi-label predictions. The agents are orchestrated with AutoGen and operate without task-specific fine-tuning, which enables parallel detection across the two-level taxonomy; this design yielded competitive results on the SemEval-2025 test set and highlights the practical zero-shot LLM systems for complex hierarchical tasks. [EN25]

2.3.2 LLMs for data augmentation

One of the most promising applications of LLMs is in addressing data scarcity and class imbalance through synthetic data generation. An LLM can be prompted to act as a domain expert and generate new text samples for underrepresented (tail) classes. This can be done by paraphrasing existing examples to increase diversity or by generating entirely new, plausible examples from scratch. This approach offers a sophisticated and flexible alternative to traditional data augmentation techniques like synonym replacement or back-translation. [CSB25; GZ24]

In more extreme cases; for example, a scenario with complete lack of labeled data, an LLM can be used to generate a large corpus of "weak" or "silver" labels. These labels, while imperfect, can serve as a starting point for training a smaller, more efficient model. Frameworks like JSDRV by Yang et al. (2024) take this a step further, using a reinforcement learning policy to select the highest-quality LLM-generated annotations for fine-tuning, creating a self-improving annotation and training pipeline [Yan+24].

2.3.3 LLMs as evaluation assistants

For extreme multi-label classification tasks with thousands of labels, manual evaluation is prohibitively expensive and time-consuming. Li et al. (2023) aimed to compare a label-ranking BiCross-Encoder against a SciBERT classifier in a very large-label setting (11,486 labels) while lacking a ready human-annotated test set. They demonstrated a practical approach by asking ChatGPT to rate the relevance of candidate labels: for each document they fed the top-10 model-predicted labels to ChatGPT and requested a 3-point relevance score (0 = irrelevant, 1 = somewhat relevant, 2 = highly relevant) along with brief explanations. The authors treated these LLM judgments as automatic annotations to compare models, then spot-checked a sample with subject-matter experts and reported about 60% agreement on the 3-point scale (rising to 82% when collapsing 1 and 2 into a single “relevant” class). They conclude that using

ChatGPT as an evaluation assistant is a cost- and time-efficient way to bootstrap evaluation when comprehensive human annotation is infeasible. [Li+23]

2.4 Focus on data-specific challenges

While architectural innovations have driven significant progress, the performance of any HMLC model is ultimately constrained by the quality and characteristics of the available data. Real-world datasets for narrative detection and other complex classification tasks are rarely pristine; they are often imbalanced, small, and increasingly, multilingual.

2.4.1 Class imbalance and tail labels

Perhaps the most pervasive challenge in real-world classification is extreme class imbalance, often described as a long-tailed distribution. In such datasets, a small number of "head" classes are represented by a vast number of training examples, while the majority of "tail" classes are represented by very few, sometimes single-digit, examples. [Hua+21] This constitutes a severe challenge for standard training algorithms, which, when optimized for overall accuracy, tend to become biased towards the majority classes and perform poorly on the infrequent but often more interesting tail classes.

The inadequation of conventional resampling

In single-label classification, a common strategy to combat imbalance is data resampling, either oversampling the minority classes or undersampling the majority classes [Luq+19; Tha+20]. However, this approach is fundamentally ill-suited for the multi-label context. An instance in an MLC dataset can simultaneously belong to multiple classes, for example, one highly frequent head class and one extremely rare tail class. If one were to oversample this instance to increase the representation of the tail class, one would inadvertently also increase the representation of the already-dominant head class, potentially exacerbating the overall imbalance with respect to other labels. This entanglement makes simple resampling ineffective and often counterproductive. [Yua+24]

A particularly effective technique that has been instancied is the distribution-based loss function. As demonstrated by Huang et al. (2021), this approach goes beyond simple re-weighting by class frequency. It introduces a loss formulation that inherently addresses both the class imbalance and the label linkage (co-occurrence) problems simultaneously. The re-weighting scheme is designed to account for the fact that an instance with multiple labels can be oversampled if each of its labels is treated independently [Hua+21].

2.4.2 Strategies for low resource and few-shot scenarios

When unlabeled data is abundant but labeling is expensive, active learning provides a principled way to build a high-quality training set with minimal human effort [Li+23]. Instead of randomly selecting data for annotation, an active learning system iteratively queries a human oracle (e.g., a domain expert) to label the instances that the model is most uncertain about or that are expected to provide the most information for improving the model. For XMC, Li et al. (2023) propose a sophisticated active learning pipeline that uses a Bi-Encoder model to first retrieve a manageable pool of potentially relevant documents for new, unlabeled classes. A greedy acquisition strategy then selects the most promising candidates from this pool for expert annotation, drastically reducing the labeling burden compared to annotating the entire corpus.[Li+24]

2.4.3 Parameter-efficient fine-tuning for low-resource adaptation

Fully fine-tuning a large PLM with billions of parameters on a small dataset is not only computationally expensive but also prone to overfitting. Parameter-Efficient Fine-Tuning (PEFT) methods have emerged as a critical technology for low-resource adaptation. [Su+24] Techniques like Adapters and Low-Rank Adaptation (LoRA) work by freezing the vast majority of the pre-trained model's weights and introducing a very small number of new, trainable parameters. LoRA, for example, injects trainable low-rank matrices into the layers of the Transformer, allowing the model to adapt to the new task by learning low-rank updates to its original weight matrices. By drastically reducing the number of trainable parameters (often by over 99%), PEFT makes the fine-tuning process more stable on small datasets, reduces memory requirements, and helps prevent catastrophic forgetting of the knowledge learned during pre-training. [Liu+24; Raz+24]

2.4.4 Data augmentation with LLMs

As discussed in Section 2.3.2, LLMs can serve as powerful data generators. For low-resource classes, an LLM can be prompted to create synthetic training examples by paraphrasing the few existing seed examples or by generating novel instances based on a class description. A cost-benefit analysis by Cegin et al. (2025) reveals that this approach provides the most significant advantage when the number of initial seed examples is extremely small (e.g., one-shot or few-shot learning) [CSB25]. As the amount of available data increases, the performance gains from LLM augmentation may diminish relative to its computational cost, and in some cases, traditional augmentation techniques can achieve comparable results more efficiently. [CSB25]

Concrete evidence of these trade-offs is provided by Whitehouse et al. (2023) [WCA23], who systematically investigate LLM-powered augmentation for multilingual commonsense reasoning. The authors use a variety of LLMs (Dollyv2, StableVicuna, ChatGPT and GPT4) to generate synthetic training examples for three low-resource benchmarks (XCOPA, XWinograd and XStoryCloze). They compare generation strategies, producing data directly in the target language, generating in English and translating the output, and using English-generated data as-is, and then fine-tune compact multilingual models (mBERT and XLMRoBERTa) on the augmented corpora.

Their results show that augmenting scarce training sets with high-quality LLM generations can substantially improve downstream accuracy (up to a 13.4point increase in the best case). GPT4 and ChatGPT produce the most natural and logically coherent examples in most languages, with GPT4 often yielding the largest gains; smaller/open models (Dollyv2, StableVicuna) deliver weaker, less consistent improvements. The authors also perform a human evaluation: native speakers rate the naturalness and logical coherence of generated examples, confirming GPT-4's superior quality but revealing languagedependent failures (notably for some lowresource languages such as Tamil). Overall, the study demonstrates that LLMs are a practical and powerful source of synthetic data for crosslingual commonsense tasks when reliable generation is available, while also highlighting limitations in coverage and the need for quality control.

In summary, the field has progressed from classical HMLC methods to sophisticated hierarchy-aware deep learning models. While recent work in the SemEval-2025 shared task has highlighted the promise of LLM-based prompting, these approaches often rely on monolithic, multi-step prompts. This paper explores a different paradigm: decomposing the HMLC task into a system of independent, specialized agents. We investigate both a parallel agentic framework and a novel actor-critic pipeline designed for enhanced traceability, directly addressing the need for more modular and reliable systems in zero-shot narrative classification.

3 Task and Data Analysis

3.1 Task description

3.1.1 SemEval2025 Task 10: Overview

SemEval-2025 Task 10 introduced a multilingual benchmark for the analysis of news narratives and entity framing across five languages and two high-impact domains (Ukraine–Russia War, Climate Change). The task is decomposed into entity framing, narrative classification, and narrative extraction subtasks, combining multi-label classification and evidence-grounded generation. Participating teams demonstrated the effectiveness of hierarchical prompting and targeted context selection; however, grounded explanation generation and label imbalance across languages remain unsolved challenges [PM25; Fan+25; SSB25a; F25].

We participated in this shared task (together with Diana Nurbakova) as competitors in Subtask 2 (Narrative Classification); the experiments reported here were carried out in that shared-task context.

This shared task asked teams to build systems that identify and extract dominant narratives from multilingual news, prioritising evidence-grounded explanations and robustness across languages. We ran multilingual evaluations to compare per-language behaviour.

3.1.2 Task decomposition

The shared task was split into three subtasks:

1. **Entity Framing (Subtask 1)**: Given an article and named-entity mentions, assign main and fine-grained role labels to each mention (multi-label, multi-class role classification). Grounding to spans and role taxonomies was required. [PM25]

2. **Narrative Classification (Subtask 2)**: Assign articles one or more narrative labels from a two-level taxonomy (main narrative → subnarrative). This is a multilingual multi-label document classification problem. [PM25]

3. **Narrative Extraction (Subtask 3)**: For a given article and its dominant narrative, generate a concise (max. 80 words) free-text explanation supporting that narrative, grounded in evidence from the article. This subtask focused on faithfulness and coverage of extracted explanations. [PM25]

3.1.3 Data and annotation

Training and evaluation data were collected from online news sources spanning 2022-mid-2024. The dataset includes articles from a mix of mainstream and lower-credibility outlets; annotations follow multilingual guidelines and predefined taxonomies for entity roles and narratives. [Sem25b]

3.1.4 Participation and headline results

The task attracted broad participation; hundreds of teams registered and dozens submitted official test results. Representative high-performing approaches included:

- **Fane et al. ("Fane")** - zero-shot LLM prompting with a hierarchical decomposition for entity framing; reported Main Role Accuracy $\approx$ 89.4% and Exact Match Ratio $\approx$ 34.5% on English. [Fan+25]

- **GateNLP** - *Hierarchical Three-Step Prompting (H3Prompt)* for Subtask 2 (narrative classification): cascade of domain → main narrative → subnarrative prompting; top performer on the English test set among competing teams. Code published. [SSB25a]

- **DEMON / TechSSN / LTG / BERTastic** - representative competitive systems using fine-tuning of large pre-trained encoders (e.g., XLM-R / DeBERTa variants) or fine-tuned LLaMA-family models for framing and narrative tasks. Several system descriptions and official proceedings papers document model design choices and per-language performance differences. [F25; P25; RN25; Nak+25]

These system papers illustrate a common trend: hierarchical or multi-step prompting and careful context selection (to fit model context windows) were effective strategies across teams. [Fan+25; SSB25a; RN25]

3.1.5 Evaluation of Subtask 2: Narrative Classification

The second subtask of SemEval-2025 Task 10 required participants to assign *narrative* (coarse-level) and *sub-narrative* (fine-level) labels to each news article. Since articles can be associated with multiple narratives, this constitutes a multi-label, multi-class document classification problem [PM25].

Evaluation target. Participants had to provide two sets of predictions per article: (1) a set of coarse-level narrative labels and (2) a set of fine-grained sub-narrative labels. Although both levels were evaluated, the official ranking was based on performance at the sub-narrative level, as this captures more nuanced framing [Sem25b].

Metrics. Two main evaluation metrics were applied:

- **Macro F1 (coarse level).** This is the unweighted mean of F1-scores computed for each coarse narrative class. It penalizes models that ignore low-frequency labels.
- **F1 (samples).** This metric is computed at the level of individual articles. For each document, precision, recall, and F1 are compute by comparing the predicted label set with the gold label set. The final score is the average of these values across all documents.

In addition, standard deviations across multiple runs and across languages were reported on the leaderboard to quantify robustness [Sem25a].

Worked example. Consider a simplified case with three possible sub-narratives: $\{N_1, N_2, N_3\}$. Suppose the gold labels for a document are $\{N_1, N_3\}$, while a system predicts $\{N_1, N_2\}$.

- **Precision** $= \frac{|\{N_1,N_2\} \cap \{N_1,N_3\}|}{|\{N_1,N_2\}|} = \frac{1}{2} = 0.5.$
- **Recall** $= \frac{|\{N_1,N_2\} \cap \{N_1,N_3\}|}{|\{N_1,N_3\}|} = \frac{1}{2} = 0.5.$
- **F1 (sample)** $= \frac{2 \times 0.5 \times 0.5}{0.5+0.5} = 0.5.$

In contrast, if the system had predicted only $\{N_1\}$, precision would be 1.0, recall 0.5, and F1 (sample) 0.67. This illustrates that partial but precise predictions may yield a higher F1 than predicting extra incorrect labels.

Implications. Because the leaderboard ranks systems by fine-level labels using sample-based F1, models must balance recall (capturing all relevant sub-narratives) with precision (avoiding irrelevant ones). Macro F1 further incentivizes attention to rare narratives, since underperformance on them disproportionately lowers the score. [PM25].

3.2 Dataset Deep Dive

We conducted a systematic descriptive analysis of the released training split to quantify its distributional, linguistic, and structural properties. Our analysis reveals several key characteristics regarding its scale, multi-label nature, class imbalance, and co-occurrence structure, which collectively motivate our architectural design choices.

Scale and Multilinguality. The corpus contains 1,699 documents in five languages: Bulgarian (BG), Portuguese (PT), English (EN), Hindi (HI), and Russian (RU). The distribution is relatively balanced across Bulgarian, Portuguese, and English ($\approx$ 399–401 documents each), with 366 Hindi documents and 133 Russian documents. This multilingual context is critical for assessing the cross-lingual generalization capabilities of classification models.

Multi-Label Nature. Building upon this multilingual foundation, a defining feature of the dataset is the high frequency of multi-label annotations. A majority of documents (54.0%) are assigned more than one narrative, with an average of 2.28 narrative labels per document and a maximum of 8. This observation confirms that multi-label modeling is essential; treating the task as a series of independent single-label classifications would discard significant structural information present in the data.

Label Frequency and Extreme Imbalance. The complexity of the multi-label structure is further compounded by severe class imbalance. The taxonomy consists of 22 distinct narrative labels and 95 sub-narrative labels, distributed in a strongly skewed manner. The most frequent narrative appears 584 times (representing $\approx 15.1\%$ of all narrative annotations in the multilingual training set), while the least frequent occurs only 9 times. This yields an imbalance ratio of approximately 65:1, creating a classic long-tailed distribution. Notably, the dataset also includes a generic Other category for documents that do not fit any predefined narrative, which accounts for a substantial portion of the data and requires models to learn a negative class boundary against all defined narratives. The most prevalent narratives are dominated by war-related topics (e.g., ‘URW: Discrediting Ukraine‘, ‘URW: Discrediting the West, Diplomacy‘), highlighting the dataset’s focus on high-stakes domains. This sparsity poses a significant challenge for supervised models, which can become biased towards majority classes and fail to generalize to rare but meaningful narratives.

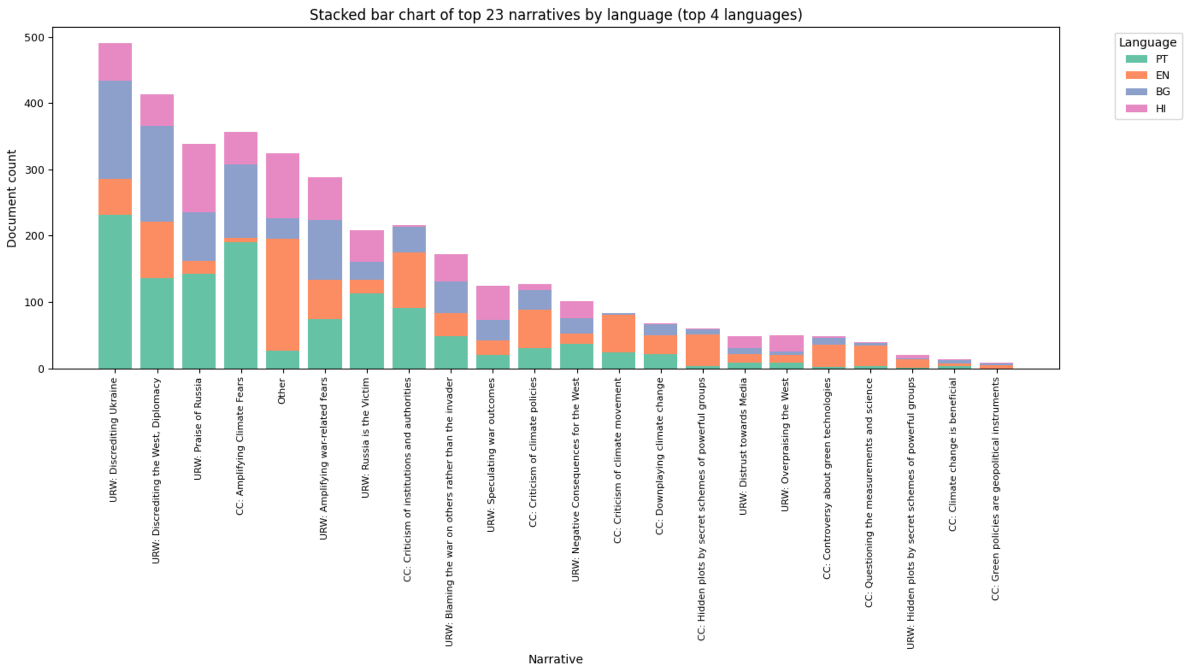


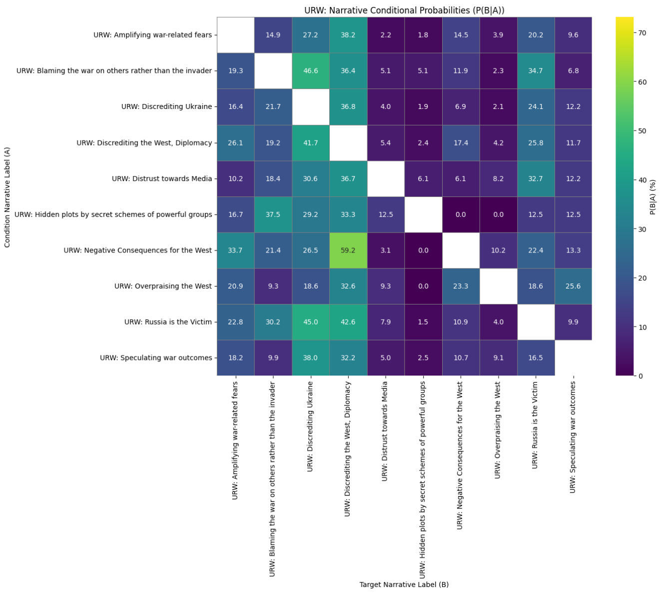
Figure 3.1: Long-tailed distribution of narrative labels in the SemEval-2025 training set. The bar chart shows the frequency of each coarse narrative, illustrating that a small number of narratives account for a large fraction of annotations while many narratives have very few examples. This severe imbalance motivates the use of zero-shot and evidence-grounded LLM approaches that do not rely on large per-class training counts.

Structured Label Co-occurrence. Despite this imbalance, the relationships between labels are not random but exhibit meaningful patterns. Co-occurrence analysis reveals that while most pairs of narratives rarely appear together, certain thematically-linked pairs exhibit strong associations. For example, documents often combine *attack* and *counter-attack* narratives (e.g., discrediting Ukraine together with discrediting the West), or pair attacks on an adversary with *praise of allies* (e.g., discrediting Ukraine alongside praise of Russia). Other recurrent patterns reflect causal framing, such as linking sanctions to economic hardship, or pairings between diplomatic failure and critiques of legitimacy. Military operations are frequently co-annotated with victim/threat framings, and conspiracy claims commonly appear together with explicit attribution to external actors. These recurring narrative couplings illustrate that co-occurrence captures structured discourse strategies in which labels reinforce one another. More advanced analysis further reveals directional dependencies (i.e., $P(B \mid A) eq P(A \mid B)$).

Language-Specific Patterns. These structural complexities are further nuanced by cross-lingual variations. The dataset exhibits notable variation across languages, with the average number of labels per document (label density) highest in Portuguese ($\approx$ 3.04) and lower in Hindi ($\approx$ 1.79), suggesting potential differences in source document complexity or annotation practices. Furthermore, the prevalence of specific narratives and their co-occurrence patterns differ across languages. This cross-lingual heterogeneity underscores the importance of using language-aware methods or, at a minimum, explicitly validating a model's performance on each language.

Document Length Heterogeneity. Finally, the dataset presents additional challenges through its variable document lengths. The corpus exhibits high variability, with a mean of approximately 404 words, a median of 371 words, and a range spanning from 45 to 4,422 words. This heterogeneity necessitates the use of effective truncation or long-text handling strategies in model design to manage computational constraints without losing critical context.

(a) CC narratives heatmap



(b) URW narratives conditional graph

Figure 3.2: Co-occurrence / conditional-probability visualizations for the most frequent narrative groups. Top: heatmap for CC narratives (normalized cell values). Bottom: conditional-probability visualization for URW narratives (directional). These paired views help identify both symmetric and directional dependencies among narratives.

4 System architectures and methods

To address the challenges of hierarchical multi-label classification presented by the SemEval-2025 dataset, we designed and evaluated three distinct architectural paradigms. We began by establishing a supervised baseline using a fine-tuned Transformer model. We then developed two zero-shot systems that leverage the advanced reasoning capabilities of Large Language Models (LLMs): a parallel agentic framework and a sequential, traceable actor-critic pipeline. This section details the design of each architecture.

4.1 Transformer baseline architecture

To establish a robust performance baseline, we first implemented a standard fine-tuning approach, which represents the conventional state-of-the-art for supervised text classification. Given the multilingual nature of the dataset, we selected XLM-RoBERTa [Con+20], a powerful multilingual encoder, as the foundation for our model.

The core architecture consists of the pre-trained XLM-RoBERTa model followed by a linear classification head. For each input text, the model produces a contextualized embedding for the special [CLS] token [**devlin_bert_2018**]. This embedding, which represents the aggregated meaning of the entire input sequence, is then passed to a single feed-forward layer that projects it into a logit vector. The dimension of this vector is equal to the total number of sub-narrative labels in the taxonomy. A sigmoid activation function is applied to these logits during inference to produce a probability between 0 and 1 for each label, allowing for multi-label predictions.

We experimented with three variations of this baseline, each designed to systematically address the specific challenges identified in our data analysis.

4.1.1 Vanilla fine-tuning

Our first implementation establishes a foundational baseline using a straightforward fine-tuning methodology. We use the XLM-RoBERTa (xlm-roberta-base) model for several reasons: the task is multilingual, and XLM-RoBERTa is pre-trained on a large multilingual corpus covering many languages, enabling robust, language-agnostic representations; moreover, as a RoBERTa-family model it benefits from improved pre-training (e.g., dynamic masking) compared to the original BERT, resulting in stronger contextual embeddings.

The architectural setup is standard: the pre-trained XLM-RoBERTa encoder is followed by a linear classification head. During a forward pass the final hidden state of the special [CLS] token is taken as an aggregate representation of the input and passed through the classification head to produce a logit vector whose dimensionality matches the number of sub-narrative labels. At inference time a sigmoid is applied to each logit to produce independent probabilities for multi-label prediction.

For this vanilla baseline we frame the problem as a set of independent binary classification tasks (one per sub-narrative) and optimize using Binary Cross-Entropy with Logits Loss (BCE-WithLogitsLoss). While simple and computationally efficient, this formulation assumes label independence and therefore does not capture structured co-occurrence patterns or hierarchical parent–child relationships present in the taxonomy; as our later variations show, explicitly addressing these properties can materially improve performance on tail and dependent labels.

Anticipated limitations. Based on our data analysis, we anticipated that this vanilla approach would face significant limitations. The primary concern is its susceptibility to the severe class imbalance. By optimizing for overall accuracy, the standard BCE loss incentivizes the model to achieve high performance on the high-frequency “head” classes, often by simply predicting “negative” for the vast majority of rare “tail” classes. This behavior is expected to result in poor performance on metrics sensitive to class imbalance, such as the macro-averaged F1 score, and serves as the primary motivation for the more advanced baselines and zero-shot architectures explored subsequently.

4.1.2 Mitigating the class imbalance

To directly address the class imbalance issue, our second variation enhances the loss function with positive class weighting. This technique modifies the standard BCE loss by increasing the penalty for misclassifying a positive instance of a rare class. For each class c we compute a per-class weight $\mathrm{pos\_weight}_c$ as the ratio of negative to positive training counts. Let N_c and P_c denote the number of negative and positive training examples for class c, respectively:

$$\mathrm{pos\_weight}_c = \frac{N_c}{P_c}.$$

This weight is then integrated directly into the BCE loss calculation, effectively increasing the cost of a false negative for rare classes. For a common label where positive and negative samples are balanced, the weight is close to 1, behaving like the standard BCE loss. However, for a rare label with a **pos_weight** of, for example, 65, the loss incurred for failing to predict that label when it is present is magnified 65 times.

This ensures that the model is more heavily penalized for failing to identify a rare narrative than for misclassifying a common one. In practice we pass the vector of per-class positive weights to the BCEWithLogitsLoss (or equivalent) implementation so rare classes contribute proportionally more to the loss. The approach is intended to improve recall on tail classes and, consequently, boost Macro F1 by forcing the model to pay closer attention to underrepresented labels.

5 Open challenges and conclusion

Despite steady progress, several critical challenges remain; addressing them will shape the next phase of research in narrative detection and hierarchical multi-label classification (HMLC).

5.1 Key open challenges

- **Interpretability and faithfulness**
  - Models must provide human-understandable explanations for predictions and demonstrate structural faithfulness (e.g., obey the true-path rule).
  - Beyond loss-based penalties, we need architectures and verification methods that guarantee or certify hierarchical consistency.
- **Dynamic and evolving hierarchies**
  - Real-world taxonomies change: new labels appear, labels split/merge, and relations shift.
  - Research directions: continual learning for label updates, modular label adapters, and lightweight schema evolution strategies that avoid full retraining.
- **Robust cross-lingual and cross-cultural transfer**
  - Transfer across typologically distant languages and across cultural framings remains unreliable.
  - Promising approaches: culture-aware adapters, contrastive cross-cultural fine-tuning, and evaluation protocols that measure cultural equivalence rather than literal translation.

- **Modeling richer narrative structure**
  - Narratives include causality, temporality, and actor-event relations that are not captured by simple taxonomies.
  - Move toward graph-structured prediction, joint event-role-narrative models, and hybrid symbolic-neural representations.
- **Governance of LLM augmentation**
  - LLM-generated training data is a powerful but noisy resource: it may amplify biases or introduce stylistic artifacts.
  - Needed tools: sampling and diversity controls, human-in-the-loop filters, bias audits, and provenance tracking for synthetic examples.

5.2 Concluding remarks

The field is shifting from purely representation-focused modelling toward systems that align pre-trained knowledge with downstream tasks (via prompts, adapters, or targeted synthetic data). Addressing interpretability, dynamics, cultural robustness, structured narratives and the governance of synthetic data will be the levers that determine whether HMLC methods are usable at scale in real-world narrative analysis.

Progress will require integrated work across evaluation, modelling, data governance, and tooling, a multidisciplinary agenda that is both technically challenging and societally important.

Bibliography

[CSB25] Jan Cegin, Jakub Simko, and Peter Brusilovsky. "LLMs vs Established Text Augmentation Techniques for Classification: When do the Benefits Outweight the Costs?" In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Luis Chiruzzo, Alan Ritter, and Lu Wang. Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 10476–10496. ISBN: 979-8-89176-189-6. DOI: 10.18653/v1/2025.naacl-long.526. URL: https://aclanthology.org/2025.naacl-long.526/ (cit. on pp. 10, 13).

[CMM11] E. A. Cherman, M. C. Monard, and J. Metz. "Multi-label Problem Transformation Methods: a Case Study". In: *CLEI Electronic Journal* 14.1 (Apr. 2011). DOI: 10.19153/cleiej.14.1.4. URL: https://doi.org/10.19153/cleiej.14.1.4 (cit. on p. 5).

[Con+20] Alexis Conneau et al. *Unsupervised Cross-lingual Representation Learning at Scale*. 2020. arXiv: 1911.02116 [cs.CL] (cit. on pp. 2, 22).

[Dev+19] Jacob Devlin et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". en. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423. URL: http://aclweb.org/anthology/N19-1423 (cit. on p. 2).

[EN25] Mohamed - Nour Eljadiri and Diana Nurbakova. “Team INSALyon2 at SemEval-2025 Task 10: A Zero-shot Agentic Approach to Text Classification”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Ed. by Sara Rosenthal et al. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 965–980. ISBN: 979-8-89176-273-2. URL: https://aclanthology.org/2025.semeval-1.129/ (cit. on p. 10).

[Fan+25] Enfa Fane et al. *Fane at SemEval-2025 Task 10: Zero-Shot Entity Framing with Large Language Models*. arXiv preprint arXiv:2504.20469. 2025. URL: https://arxiv.org/abs/2504.20469 (cit. on pp. 15–17).

[F25] M. Fenu and ... “DEMON at SemEval-2025 Task 10: Fine-tuning LLaMA-3 for Multilingual Entity Framing”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.192/ (cit. on pp. 15, 16).

[GZ24] Anna V. Glazkova and Olga V. Zakharova. *Evaluating LLM Prompts for Data Augmentation in Multi-label Classification of Ecological Texts*. 2024. arXiv: 2411.14896 [cs.CL]. URL: https://arxiv.org/abs/2411.14896 (cit. on p. 10).

[Gon+20] Jibing Gong et al. “Hierarchical Graph Transformer-Based Deep Learning Model for Large-Scale Multi-Label Text Classification”. In: *IEEE Access* 8 (2020), pp. 30885–30896. DOI: 10.1109/ACCESS.2020.2973121 (cit. on p. 2).

[Hu+25] W. Hu et al. “A Survey of Multi-Label Text Classification Under Few-Shot Scenarios”. In: *Applied Sciences* 15 (2025), p. 8872. DOI: 10.3390/app15168872. URL: https://doi.org/10.3390/app15168872 (cit. on p. 1).

[Hua+24] S. Huang et al. “Application of Label Correlation in Multi-Label Classification: A Survey”. In: *Applied Sciences* 14 (2024), p. 9034. DOI: 10.3390/app14199034. URL: https://doi.org/10.3390/app14199034 (cit. on p. 1).

[Hua+21] Yi Huang et al. “Balancing Methods for Multi-label Text Classification with Long-Tailed Class Distribution”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens et al. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 8153–8161. DOI: 10.18653/v1/2021.emnlp-main.643. URL: https://aclanthology.org/2021.emnlp-main.643/ (cit. on pp. 11, 12).

[Li+23] Dan Li et al. “Enhancing Extreme Multi-Label Text Classification: Addressing Challenges in Model, Data, and Evaluation”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Ed. by Mingxuan Wang and Imed Zitouni. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 313–321. DOI: 10.18653/v1/2023.emnlp-industry.30. URL: https://aclanthology.org/2023.emnlp-industry.30/ (cit. on pp. 11, 12).

[Li+24] Dongyuan Li et al. *A Survey on Deep Active Learning: Recent Advances and New Frontiers*. 2024. arXiv: 2405.00334v2 [cs.LG]. URL: https://arxiv.org/abs/2405.00334v2 (cit. on p. 12).

[Li+21] Irene Li et al. “Heterogeneous Graph Neural Networks for Multi-label Text Classification”. In: *Lecture Notes in Computer Science*. arXiv:2103.14620. 2021. DOI: 10.48550/ARXIV.2103.14620. URL: https://arxiv.org/abs/2103.14620 (cit. on p. 8).

[LDH24] Xinyu Li, Jiaman Ding, and Shuang Hu. “Relative Entropy and PageRank-Based Classifier Chains for Multi-Label Classification”. In: *IEEE Access* 12 (2024), pp. 87665–87674. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2024.3409714. URL: https://ieeexplore.ieee.org/document/10549888/ (cit. on p. 2).

[Liu+23] Rundong Liu et al. *Recent Advances in Hierarchical Multi-label Text Classification: A Survey*. 2023. arXiv: 2307.16265 [cs.CL]. URL: https://arxiv.org/abs/2307.16265 (cit. on pp. 1, 2).

[Liu+24] Zequan Liu et al. “ALoRA: Allocating Low-Rank Adaptation for Fine-tuning Large Language Models”. In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Long Papers*. Mexico City, Mexico, June 2024, pp. 622–641. DOI: 10.18653/v1/2024.naacl-long.35. URL: https://aclanthology.org/2024.naacl-long.35 (cit. on p. 12).

[Luq+19] Amalia Luque et al. “The impact of class imbalance in classification performance metrics based on the binary confusion matrix”. In: *Pattern Recognit.* 91 (2019), pp. 216–231. DOI: 10.1016/J.PATCOG.2019.02.023 (cit. on p. 11).

[Nak+25] Preslav Nakov et al. “BERTastic at SemEval-2025 Task 10: Fine-tuned Transformers for Multilingual Narrative Analysis”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.177/ (cit. on p. 16).

[Pen+21] Hao Peng et al. “Hierarchical Taxonomy-Aware and Attentional Graph Capsule RCNNs for Large-Scale Multi-Label Text Classification”. In: *IEEE Transactions on Knowledge and Data Engineering* 33.6 (2021), pp. 2505–2519. DOI: 10.1109/TKDE.2019.2956257 (cit. on p. 2).

[PM25] Jakub Piskorski, Tarek Mahmoud, and ... “SemEval-2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. ACL Anthology. 2025. URL: https://aclanthology.org/2025.semeval-1.331/ (cit. on pp. 15–18).

[Pis+25] Jakub Piskorski et al. “SemEval-2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval 2025)*. Vienna, Austria, 2025 (cit. on p. 1).

[P25] P. Premnath and ... “TECHSSN at SemEval-2025 Task 10: A Comparative Study for Narrative Extraction”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.286/ (cit. on p. 16).

[Raz+24] Olesya Razuvayevskaya et al. “Comparison between parameter-efficient techniques and full fine-tuning: A case study on multilingual news article classification”. In: *PLOS ONE* 19.5 (2024), e0301738. DOI: 10.1371/journal.pone.0301738. URL: https://doi.org/10.1371/journal.pone.0301738 (cit. on p. 12).

[Rea+11] J. Read et al. “Classifier Chains for Multi-label Classification”. In: *Machine Learning* 85.3 (2011), pp. 333–359. DOI: 10.7007/s10994-011-5256-5. URL: https://doi.org/10.1007/s10994-011-5256-5 (cit. on p. 5).

[Rea+21] J. Read et al. “Classifier Chains: A Review and Perspectives”. In: *Journal of Artificial Intelligence Research* 70 (2021), pp. 683–718. DOI: 10.1613/jair.1.12376. URL: https://doi.org/10.1613/jair.1.12376 (cit. on p. 5).

[RFR22] Miguel Romero, Jorge Finke, and Camilo Rocha. “A top-down supervised learning approach to hierarchical multi-label classification in networks”. In: *Applied Network Science* 7.1 (2022), p. 8. DOI: 10.1007/s41109-022-00445-3. URL: https://doi.org/10.1007/s41109-022-00445-3 (cit. on p. 7).

[RN25] Egil Rønningstad and Gaurav Negi. *LTG at SemEval-2025 Task 10: Optimizing Context for Classification of Narrative Roles*. arXiv preprint arXiv:2506.05976. 2025. URL: https://arxiv.org/abs/2506.05976 (cit. on pp. 16, 17).

[SC22] Mobashir Sadat and Cornelia Caragea. "Hierarchical Multi-Label Classification of Scientific Documents". en. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, 2022, pp. 8923–8937. DOI: 10.18653/v1/2022.emnlp-main.610. URL: https://aclanthology.org/2022.emnlp-main.610 (cit. on p. 2).

[Sem25a] SemEval-2025 Task 10 Organizers. *SemEval 2025 Task 10 Leaderboard (official results)*. https://propaganda.math.unipd.it/semeval2025task10/leaderboard.php. Accessed 2025-09-16. 2025 (cit. on p. 17).

[Sem25b] SemEval-2025 Task 10 Organizers. *SemEval 2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News (task webpage)*. https://propaganda.math.unipd.it/semeval2025task10/. Accessed 2025-09-16. 2025 (cit. on pp. 16, 17).

[Sha+18] Jincheng Shan et al. "Co-learning Binary Classifiers for LP-Based Multi-label Classification". en. In: *Intelligence Science and Big Data Engineering*. Ed. by Yuxin Peng et al. Vol. 11266. Cham: Springer International Publishing, 2018, pp. 443–453. ISBN: 9783030026974 9783030026981. DOI: 10.1007/978-3-030-02698-1_39. URL: https://link.springer.com/10.1007/978-3-030-02698-1_39 (cit. on p. 2).

[SSB25a] Iknoor Singh, Carolina Scarton, and Kalina Bontcheva. *GateNLP at SemEval-2025 Task 10: Hierarchical Three-Step Prompting for Multilingual Narrative Classification*. arXiv preprint arXiv:2505.22867. 2025. URL: https://arxiv.org/abs/2505.22867 (cit. on pp. 2, 15–17).

[SSB25b] Iknoor Singh, Carolina Scarton, and Kalina Bontcheva. *GateNLP H3Prompt System Code Repository*. https://github.com/GateNLP/semeval2025-task10. GitHub repository for hierarchical three-step prompting system. 2025 (cit. on p. 2).

[Su+24] Tong Su et al. "Unlocking Parameter-Efficient Fine-Tuning for Low-Resource Language Translation". In: *Findings of the Association for Computational Linguistics: NAACL 2024*. Ed. by Kevin Duh, Helena Gomez, and Steven Bethard.

Mexico City, Mexico: Association for Computational Linguistics, June 2024, pp. 4217–4225. DOI: 10.18653/v1/2024.findings-naacl.263. URL: https://aclanthology.org/2024.findings-naacl.263/ (cit. on p. 12).

[Suc+14] L. Sucar et al. "Multi-label classification with Bayesian network-based chain classifiers". In: *Pattern Recognit. Lett.* 41 (2014), pp. 14–22. DOI: 10.1016/J.PATREC.2013.11.007 (cit. on pp. 4, 5).

[Tha+20] F. Thabtah et al. "Data imbalance in classification: Experimental evaluation". In: *Inf. Sci.* 513 (2020), pp. 429–441. DOI: 10.1016/j.ins.2019.11.004 (cit. on p. 11).

[TS18] Vaishali S. Tidake and Shirish S. Sane. "Multi-label Classification: a survey". In: *International Journal of Engineering & Technology* 7.4.19 (2018), p. 1045. DOI: 10.14419/ijet.v7i4.19.28284. URL: https://doi.org/10.14419/ijet.v7i4.19.28284 (cit. on pp. 1, 2).

[Wan+23] Fengjun Wang et al. "Text2Topic: Multi-Label Text Classification System for Efficient Topic Detection in User Generated Content with Zero-Shot Capabilities". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Ed. by Mingxuan Wang and Imed Zitouni. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 93–103. DOI: 10.18653/v1/2023.emnlp-industry.10. URL: https://aclanthology.org/2023.emnlp-industry.10/ (cit. on p. 9).

[WDH24] Kunze Wang, Yihao Ding, and Soyeon Caren Han. *Graph Neural Networks for Text Classification: A Survey*. 2024. DOI: https://doi.org/10.1007/s10462-024-10808-0. arXiv: 2304.11534 [cs.CL]. URL: https://arxiv.org/abs/2304.11534 (cit. on p. 8).

[WCB18] Jonatas Wehrmann, Ricardo Cerri, and Rodrigo Barros. "Hierarchical Multi-Label Classification Networks". In: *Proceedings of the 35th International Conference on Machine Learning*. Ed. by Jennifer Dy and Andreas Krause. Vol. 80. Proceedings of Machine Learning Research. PMLR, July 2018, pp. 5075–5084. URL: http://proceedings.mlr.press/v80/wehrmann18a.html (cit. on p. 7).

[Wen+20] Wei Weng et al. "Label Specific Features-Based Classifier Chains for Multi-Label Classification". In: *IEEE Access* 8 (2020), pp. 51265–51275. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2020.2980551. URL: https://ieeexplore.ieee.org/document/9035463/ (cit. on p. 2).

[WCA23] Chenxi Whitehouse, Monojit Choudhury, and Alham Fikri Aji. *LLM-powered Data Augmentation for Enhanced Cross-lingual Performance*. 2023. arXiv: 2305.14288 [cs.CL]. URL: https://arxiv.org/abs/2305.14288 (cit. on p. 13).

[Xu+21] Linli Xu et al. "Hierarchical Multi-label Text Classification with Horizontal and Vertical Category Correlations". In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens et al. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2459–2468. DOI: 10.18653/v1/2021.emnlp-main.190. URL: https://aclanthology.org/2021.emnlp-main.190/ (cit. on p. 6).

[Yan+24] Ruichao Yang et al. "Reinforcement Tuning for Detecting Stances and Debunking Rumors Jointly with Large Language Models". In: *Findings of the Association for Computational Linguistics: ACL 2024*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 13423–13439. DOI: 10.18653/v1/2024.findings-acl.796. URL: https://aclanthology.org/2024.findings-acl.796/ (cit. on p. 10).

[Yua+24] Ling Yuan et al. "Research of Multi-Label Text Classification based on Label Attention and Correlation Networks". In: *PLOS ONE* 19.9 (2024), e0311305. DOI: 10.1371/journal.pone.0311305. URL: https://doi.org/10.1371/journal.pone.0311305 (cit. on p. 11).

[Zan+24] A. Zangari et al. "Hierarchical Text Classification and Its Foundations: A Review of Current Research". In: *Electronics* 13.7 (2024), p. 1199. DOI: 10.3390/electronics13071199. URL: https://doi.org/10.3390/electronics13071199 (cit. on pp. 1, 6).

[Zha+18a] M.-L. Zhang et al. "Binary relevance for multi-label learning: an overview". In: *Frontiers of Computer Science* 12.2 (2018), pp. 191–202. DOI: 10.1007/s11704-017-7031-7. URL: https://doi.org/10.1007/s11704-017-7031-7 (cit. on p. 5).

[Zha+18b] Min-Ling Zhang et al. "Binary relevance for multi-label learning: an overview". en. In: *Frontiers of Computer Science* 12.2 (Apr. 2018), pp. 191–202. ISSN: 2095-2228, 2095-2236. DOI: 10.1007/s11704-017-7031-7. URL: http://link.springer.com/10.1007/s11704-017-7031-7 (cit. on pp. 2, 4).

[Zho+20] Jie Zhou et al. "Hierarchy-Aware Global Model for Hierarchical Text Classification". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Ed. by Dan Jurafsky et al. Online: Association for Computational Linguistics, July 2020, pp. 1106–1117. DOI: `10.18653/v1/2020.acl-main.104`. URL: `https://aclanthology.org/2020.acl-main.104/` (cit. on pp. 7, 8).

Eidesstattliche Erklärung

Hiermit versichere ich, dass ich diese Masterarbeit selbstständig und ohne Benutzung anderer als der angegebenen Quellen und Hilfsmittel angefertigt habe und alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, als solche gekennzeichnet sind, sowie, dass ich die Masterarbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

Passau, September 17, 2025

Nour Eljadiri